Solutions to Review Questions for Lecture 12

1. This will be $10! = 3,626,800$, which is closer to one million.
2. This is $15! = 1.3 * 10^{12}$, which is closer to one trillion.
3. This would be $\binom{10}{5} = 10 * 9 * 8 * 7 * 6/5! = 252$, which is closer to one hundred.
4. This would be $\binom{25}{10} = 25!/(10! * 15!) = 3,268,760$, which is closer to one million.
5. This is equivalent to choosing which 4 of the 11 positions contain an "A", and so the number of ways is $\binom{11}{4} = 11 * 10 * 9 * 8/4! = 330$.
6. This is equivalent to choosing which 4 of the 14 positions contain an "A" and which 7 of the 10 remaining positions contain a "B". Each of the configurations with 4 "A" 's will have as possible all of the configurations with 7 "B" 's, and so the total number of ways is $\binom{14}{4} * \binom{10}{7} = \frac{14!*10!}{4!*10!*7!*3!} = \frac{14!}{4!*7!*3!} = 120,120$.
7. This will be the binomial probability $\binom{7}{3}(12/20)^3(8/20)^4 = 0.19$.
8. This will be the hypergeometric probability $\frac{\binom{12}{4}\binom{8}{3}}{\binom{20}{7}} = 0.36$.
9. a) This will be $\frac{\binom{1}{1}\binom{29}{2}}{\binom{30}{3}} = 0.1$.
   b) This will be $1 - P(none) = 1 - \frac{\binom{3}{0}\binom{27}{3}}{\binom{30}{3}} = 0.28$.
   c) This will be $\frac{\binom{3}{2}\binom{27}{1}}{\binom{30}{3}} = 0.020$.
10. This will be $\frac{\binom{350}{35}\binom{400}{40}\binom{250}{25}}{\binom{1,000}{100}}$.
11. a) There are 36 possibilities for the first two die rolls and 6 of them have the first two die rolls the same. Hence the probability is $6/36 = 1/6$.
    b) The chance of a 6 is $1/6$ and this is a binomial experiment. Hence the probability will be $\binom{7}{2}(1/6)^2(5/6)^5 = 0.23$.
    c) This is equivalent to the probability of having exactly 1 6 in the last 3 rolls, which will be $\binom{3}{1}(1/6)^1(5/6)^2 = 0.35$.
    d) This is $P(3\ sixes|at\ least\ 1\ six) = \frac{P(3\ sixes)}{P(at\ least\ 1\ six)} = \frac{P(3\ sixes)}{1-P(0\ sixes)}$. This evaluates to $\frac{\binom{7}{3}(1/6)^3(5/6)^4}{1-(5/6)^7} = 0.11$.
12. a) There are 198 people with high school or less education, 382 people with at least some college, and 580 people total. Hence this will be $\frac{\binom{198}{100}\binom{382}{180}}{\binom{580}{280}}$.
    b) This will be $\sum_{k=100}^{198} \frac{\binom{198}{k}\binom{382}{280-k}}{\binom{580}{280}}$.
    c) The answer to part a) is `dhyper(100,198,382,280)=0.052` and the answer to part b) is `1-phyper(99,198,382,280)=0.25`. Note that `phyper` uses 99 because 100 is part of the sum in part b).
    d) No, a probability of 25% is not particularly unusual if education and partisanship are unrelated.